

BUSINESS REQUIREMENTS DOCUMENT

Women's Interest-Free Loan Management System (WILMS)

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1. Executive Summary

The Women's Interest-Free Loan Management System (WILMS) is a purpose-built digital platform designed to replace manual, paper-based loan management with a structured, transparent, and fully automated system for managing interest-free micro-loan programs targeting women across multiple communities.

The system enforces strict repayment rules, provides real-time financial tracking, supports field collectors with a simple mobile-first interface, and produces accurate financial reports — eliminating the risks of manual record-keeping, collector fraud, and data inconsistency.

WILMS is built on a transaction-first, role-based architecture where all balances, statuses, and reports are derived purely from verified transaction records.

2. Document Scope & Purpose

2.1 Purpose

This Business Requirements Document (BRD) defines the functional and non-functional requirements for WILMS. It provides a definitive reference for the development team, stakeholders, and auditors.

2.2 Scope

- This document covers:
- Borrower registration, verification, and approval workflows
 - Loan disbursement and repayment cycle management
 - Group and collector management structures
 - Automated tracking, defaults, and risk flagging
 - Financial reconciliation and reporting
 - Security, audit, and fraud prevention controls
 - SMS/Email communication to borrowers
 - Edge case handling and enforcement rules

2.3 Out of Scope

- Interest-bearing loan products
- Borrower-facing self-service portal (Phase 2)
- Direct bank/mobile money integrations (Phase 2)

3. Stakeholders & User Roles

WILMS is role-based. Each role has defined permissions and restrictions. The table below summarises all system roles.

Role	Permissions	Restrictions
Super Admin	Full system control. Manage all users, loans, withdrawals, configurations, and audit logs.	None — highest privilege level.
Collector	Register borrowers; record payments and admin fees; manage assigned communities/groups; view own performance.	Cannot edit past records. Cannot access system-wide financials. Same-day edits only.
Registration Officer	Register borrowers; upload documents and photos; view own registered borrowers.	No approvals. No payments. No financial access.
Approver	Verify and approve or reject borrower applications. Can be assigned to a collector.	Cannot disburse funds or record payments.

4. Borrower Registration

4.1 Required Profile Data

Category	Fields	Notes
Personal Details	Full Name, Date of Birth, Gender, Phone Number, Email, Nationality, ID Type, ID Number	Phone number must be unique. Email optional.
Address	House Address, GPS Address, City, Region, District	Full address required for field visits.
Business Info	Business Name, Business Address, Type of Work	Required to assess repayment capacity.
Guarantor	Name, Phone, Relationship	Mandatory. Guarantor is fully liable for repayment.
Photo	Passport-style photo upload	Mandatory. Used for identity verification.

4.2 Automatic Duplicate & Conflict Detection

Before saving a registration, the system automatically checks:

- Duplicate phone numbers across all borrowers
- Duplicate ID numbers (same type and number)
- Similar name matching (fuzzy match to surface near-duplicates)
- Whether the applicant already holds an active loan
- Blacklist status — blocked if previously blacklisted

If any check fails, registration is blocked and the reason is displayed to the Registration Officer. All flagged attempts are logged.

5. Verification & Approval Workflow

5.1 Approval States

Status	Description
Pending	Default state after registration. Awaiting Approver review.
Approved	Borrower cleared. Eligible for loan disbursement (after admin fee payment).
Rejected	Application declined. Reason logged. Borrower may re-apply after review.
Blacklisted	Permanently barred. Cannot register, receive loans, or be guarantor.

All approval actions (Approve / Reject / Blacklist) are permanently logged with the acting user, timestamp, and reason.

6. Administration Fee

- Every approved borrower must pay an administration fee before loan disbursement.
- The collecting Collector's name is automatically recorded against the fee transaction.
- Loan disbursement is system-blocked until the fee transaction is confirmed.
- The fee is logged as a separate transaction type (Admin Fee) in the financial ledger.

7. Group Structure & Joint Liability

7.1 Group Rules

- All borrowers are organised into groups within their community.
- Each group elects a group leader who serves as a point of contact.
- All group members are jointly liable for each other's repayments.
- If one member defaults, the group is expected to cover the shortfall.
- Repeated group-level defaults result in the group being flagged at risk.

7.2 Group Risk Levels

Level	Trigger Condition
Low Risk	All members paying consistently. No missed payments.
At Risk	One or more members have missed payments in the current cycle.
Flagged	Multiple members in default or repeated missed payment events.
Suspended	Group barred from new loans until outstanding obligations are cleared.

8. Loan Structure

8.1 Loan Configuration

Each loan created in the system contains the following parameters:

Parameter	Description
Loan Amount	Principal disbursed to the borrower. No interest applied.
Loan Duration	Total number of weeks in the repayment cycle (e.g., 20 weeks).
Weekly Payment Amount	Fixed amount due each week. $\text{Loan Amount} \div \text{Duration}$.
Payment Day	Assigned day of the week for all repayments.
Loan Cycle / Batch	Identifies the disbursement batch for grouped reporting.
Start Date	First repayment date; determines the full weekly schedule.

9. Payment Rules & Enforcement

9.1 Core Payment Rules

- All payments are weekly and must be made on the assigned payment day.
- Full weekly amount must be paid — no partial payments are accepted.
- Advance payments (paying ahead of schedule) are not permitted.
- Payments must always clear the oldest unpaid obligation first.
- The system cannot record less than the required amount for any week.

9.2 Missed Payment Handling

When a payment week is missed:

- The week is automatically marked as Missed in the schedule.
- The unpaid amount is carried forward to the next payment week.
- The next required payment = Current week amount + All outstanding missed amounts.
- This accumulates until all arrears are cleared.

9.3 Payment Edit Controls

- Same-day corrections are permitted for Collectors only.
- After the payment day ends, all payment records are permanently locked.
- Locked records can only be corrected via a formal Adjustment transaction, which requires Supervisor (Super Admin) approval.
- All edits — including same-day changes — are logged with timestamp, user, and reason.

10. Automated Loan Tracking System

All loan tracking is fully automated. No manual balance calculations are required or permitted.

10.1 Automatic Balance Calculation

For each borrower, in real time:

- Total Paid = sum of all confirmed repayment transactions
- Remaining Balance = Loan Amount — Total Paid
- Overpayment is flagged automatically and queued for review

10.2 Weekly Schedule Generation

- Upon loan creation, the system generates a complete weekly schedule from Week 1 to Week N.
- Each week has an assigned date, expected amount, and status (Pending / Paid / Missed).
- Statuses update automatically as transactions are recorded.

10.3 Defaulter Detection

Borrower Status	Trigger	Action
Active	All payments on track.	No action.
At Risk	1 missed payment in current cycle.	Flag borrower. Notify Collector via alert.
Defaulted	2 or more consecutive missed payments.	Flag borrower. Escalate to Supervisor. Notify guarantor.
Blacklisted	Court/recovery action or voluntary blacklist by Admin.	Block from all future loans.

10.4 Real-Time Loan Progress Metrics

The borrower profile automatically displays:

- % of loan repaid
- Weeks completed vs weeks remaining
- Payment consistency score
- Total missed payments in the current cycle

11. Financial System

11.1 Transaction-Only Ledger

Every financial movement in WILMS is recorded as a transaction. No manual balance entries are permitted. All reports and dashboards are derived from the transaction ledger.

11.2 Transaction Types

Type	Description	Who Records
Disbursement	Loan principal paid to borrower.	Super Admin / Collector (if permitted)
Repayment	Weekly loan repayment from borrower.	Collector
Admin Fee	One-time fee before disbursement.	Collector
Withdrawal	Funds withdrawn from the loan pool.	Super Admin only
Adjustment	Correction to a locked transaction.	Super Admin only (with audit log)

12. Daily Reconciliation

At the end of each collection day, Collectors must complete a reconciliation:

Reconciliation Field	Description
Expected Total	System-calculated sum of all payments due today across assigned borrowers.
Collected Total	Sum of repayments recorded in the system by the Collector today.
Physical Cash	Actual cash counted by the Collector and entered manually.
Variance	System flags any discrepancy between Collected Total and Physical Cash.
Supervisor Approval	Variances above a defined threshold require Super Admin review.

13. Fraud Prevention & Audit Controls

13.1 Known Fraud Risks

The following collector-level fraud risks have been identified and are addressed by the system:

Risk	Description	Mitigation
Fake Repayment Entry	Collector records payments not actually collected.	Mandatory daily reconciliation. Physical cash vs system must match.
Unrecorded Cash Collection	Collector collects but does not enter in system.	Borrower SMS receipts confirm every payment. Discrepancy flagged.
Malicious Same-Day Edits	Collector alters records within the allowed same-day window.	All edits are timestamped and audit-logged. Supervisor notified.
Duplicate Registration	Collector registers the same person twice for multiple loans.	Automated duplicate detection on phone, ID, and name.

13.2 Audit & Location Controls

- GPS coordinates and timestamps are automatically captured for every payment transaction recorded in the field.
- Location data is stored with each record and visible in the audit log.
- Any same-day edit triggers an automatic alert to the Supervisor.
- All adjustments require Super Admin approval before taking effect.
- The full audit log is immutable — no records can be deleted.

14. SMS & Email Communication

14.1 Purpose

To ensure borrower transparency, reduce disputes, and deter collector fraud, WILMS sends automated notifications to borrowers at key events.

14.2 Notification Events

Event	Channel	Message Summary
Registration Approved	SMS + Email	Your loan application has been approved. Please pay the administration fee to proceed.
Registration Rejected	SMS	Your application was not approved at this time. Contact your loan officer for details.
Loan Disbursed	SMS + Email	GHS [amount] has been disbursed to you on [date]. Your first payment is due on [date].
Payment Received	SMS	Payment of GHS [amount] received on [date]. Remaining balance: GHS [balance]. Thank you.
Payment Reminder	SMS	Reminder: Your payment of GHS [amount] is due on [date]. Please ensure timely payment.
Missed Payment	SMS	A payment was missed on [date]. Your next required payment is GHS [catchup amount].
Defaulter Status	SMS + Email	Your account has been flagged as Defaulted. Please contact your loan officer urgently.
Loan Completed	SMS + Email	Congratulations! Your loan has been fully repaid. You are eligible for a new loan cycle.
Guarantor Alert	SMS	This is to inform you that the borrower you guaranteed has missed a payment on [date].

SMS is the primary channel given field conditions. Email is supplementary where borrower email is registered. Both channels are templated and automatically triggered — no manual sending is required.

15. Collector Performance Tracking

The system automatically calculates and displays each Collector's performance:

Metric	Description
Expected Collections	Total amount due from all borrowers assigned to this Collector today.
Actual Collections	Total amount recorded in the system by the Collector.
Collection Rate	Actual ÷ Expected expressed as a percentage.
Missed Borrowers	Number of borrowers who missed payment and were not followed up.
Reconciliation Variance	Daily discrepancy between recorded and physical cash.
Edit Frequency	Number of same-day edits made — high frequency triggers review.

16. Edge Cases & System Responses

Edge Case	Scenario	System Response
Death of Borrower	Borrower passes away mid-cycle.	Loan flagged. Guarantor notified. Supervisor manually reviews for write-off or recovery.
Guarantor Unreachable	Guarantor cannot be contacted after default.	Record flagged. Escalated to Super Admin for legal/recovery action.
Overpayment	Collector records amount exceeding the required payment.	Overpayment flagged. Blocked by system. Requires review and adjustment.
Public Holiday / Day Off	Payment day falls on a holiday.	Super Admin reschedules the affected payment day. System updates schedule automatically.
Borrower Relocates	Borrower moves out of assigned community.	Collector updates profile. Super Admin may reassign to another Collector/group.
System Downtime	System unavailable during collection.	Offline mode captures data locally; syncs when connectivity resumes.
Duplicate Transaction	Same payment entered twice by Collector.	System detects duplicate amount + date + borrower. Blocks second entry and alerts Supervisor.
Group Dissolution	All members of a group default or leave.	Group marked Suspended. Outstanding loans remain with individual borrowers.

Loan Write-Off	Unrecoverable loan after all recovery attempts.	Super Admin records a Write-Off adjustment. Borrower blacklisted. Full audit trail retained.
Wrong Loan Amount Disbursed	Incorrect amount was disbursed.	Disbursement locked after same day. Super Admin raises adjustment with documentation.

17. Dashboard & Reports

17.1 System Dashboard (Super Admin)

- Total funds in the loan pool
- Total amount disbursed
- Total amount collected to date
- Total outstanding balances
- Number of active, at-risk, and defaulted borrowers
- Collector performance summary
- Group risk distribution

17.2 Collector Dashboard

- My borrowers — payment status today
- Expected vs collected for today
- Missed payment alerts
- Reconciliation status

17.3 Standard Reports

Report Name	Description
Loan Portfolio Report	Full view of all active loans, balances, and statuses.
Daily Collection Report	Payments collected per day, per Collector.
Defaulter Report	All borrowers flagged as At Risk or Defaulted.
Collector Performance Report	Collection rates, variances, and edit logs by Collector.
Group Risk Report	Risk levels and missed payments by group.
Financial Ledger Report	Full transaction log for any date range.
Audit Log Report	All system actions by user, with timestamps.

18. Non-Functional Requirements

Category	Requirement	Target
Performance	Page load time for Collector dashboard.	< 2 seconds on 3G network.
Availability	System uptime.	99.5% monthly.
Offline Mode	Collector can capture payments offline.	Sync within 15 minutes of reconnection.
Security	All sensitive data encrypted at rest and in transit.	AES-256 / TLS 1.2+.
Audit Logs	All transactions and edits logged immutably.	Retained for 7 years.
Scalability	Support concurrent field collections.	Up to 500 concurrent users.
SMS Delivery	Payment confirmation SMS delivered to borrower.	Within 60 seconds of recording.
Backup	Automated daily database backup.	Recovery point objective < 24 hours.

19. Security & Data Privacy

- Role-based access control (RBAC) — each user sees only what their role permits.
- Sensitive fields (ID numbers, phone numbers, GPS addresses) are encrypted at rest.
- All API calls require authenticated sessions with token expiry.
- Audit logs capture every read, write, and delete action with user identity and timestamp.
- Data backup is automated daily; recovery tested quarterly.
- Borrower personal data is handled in accordance with applicable data protection regulations.
- GPS data collected during field operations is used for fraud detection only and not shared.

20. Assumptions & Dependencies

20.1 Assumptions

- Collectors have access to a smartphone with a data connection (or periodic WiFi).
- A licensed SMS gateway provider will be contracted and configured before go-live.
- Super Admin will configure loan rules, cycles, and community structures before the first disbursement.
- Borrowers have valid phone numbers that are reachable for SMS delivery.
- All financial figures are in Ghanaian Cedis (GHS) unless otherwise configured.

20.2 Dependencies

- SMS gateway integration (e.g., Hubtel, Arkesel, or equivalent).
- Email service provider (e.g., SendGrid or equivalent).
- Mobile device GPS availability for location-stamped transactions.

- Cloud hosting with 99.5% uptime SLA.

21. Glossary

Term	Definition
WILMS	Women's Interest-Free Loan Management System.
BRD	Business Requirements Document.
Collector	Field officer responsible for registering borrowers and collecting repayments.
Approver	Role responsible for verifying and approving borrower applications.
Guarantor	Third party who assumes full liability if the borrower fails to repay.
Loan Cycle	A defined period and batch of loans disbursed together.
Adjustment	A Super Admin-approved correction to a locked financial record.
Reconciliation	End-of-day process to match recorded collections with physical cash.
Blacklist	Permanent ban from the loan program due to default or fraud.
At Risk	Borrower or group with one or more missed payments in the current cycle.
Defaulted	Borrower with two or more consecutive missed payments.
GPS Address	Digital location identifier used to verify borrower and collector location.
Admin Fee	One-time fee charged before loan disbursement, not deducted from the loan.

22. Document History & Sign-Off

22.1 Version History

Version	Date	Author	Change Summary
1.0	May 2026	Business Analyst Team	Initial release

22.2 Sign-Off

Name & Role	Signature	Date
Project Sponsor		
Product Owner		
Lead Developer		
Operations Manager		

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